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Prediction of waste generation forecast and emission potential on the Erode City solid waste dump yards based on machine learning approach

E. B. Priyanka[1], S. Vijayashanthy[2], S. Thangavel[1], R. Anand[3,4]✉, G. B. Bhavana[3], Baseem Khan[4,5,6]✉, K. Jeyanthi[7] & A. Ambikapathy[8]

Proposed research presents a data-driven framework for forecasting municipal solid waste (MSW) generation and emission dynamics in Erode City, India, by employing supervised machine learning algorithms. Leveraging a five-year dataset (2019–2024) comprising socio-economic variables, zonal waste typologies, and historical waste volumes, the model integrates Support Vector Machine (SVM), Random Forest (RF), and Naive Bayes (NB) classifiers. Feature selection and proximity ranking techniques were applied to identify high-impact variables, with plastic and organic waste emerging as dominant predictors. Data pre-processing included normalization, missing value imputation, and spatial zoning analysis. The model was validated through cross-validation with an 80:20 training-to-testing ratio. Among the tested models, SVM exhibited Superior performance, achieving a prediction accuracy of 96%, lowest mean squared error (MSE = 4860), and minimal computational latency (0.67 seconds), indicating suitability for real-time deployment. The integration of proximity matrix analysis and zonal feature clustering enhanced interpretability and robustness. The proposed framework demonstrates significant potential for scalable waste forecasting applications, enabling emission quantification and strategic decision-making. Future work includes the incorporation of real-time sensor data, temporal decomposition, and hybrid deep learning architectures to optimize waste handling and carbon mitigation strategies.

Keywords Solid Waste Management, Machine Learning, SVM, RF, NB

Any items that are in an intact condition and are wasted, worthless, or undesirable are referred to as solid waste (SW). The vast majority of SW is produced by commercial, institutional, residential, and industrial operations. This can include a variety of things, including waste from the home, construction waste, packaging, and technological waste. Because inappropriate disposal can have detrimental effects on the environment, human health, and general well-being, SW management is a crucial environmental concern[1]. Acquisition, transit, destruction, composting, and SW reduction are all essential elements of effective SW management. Gathering and moving SW from its source to a specified disposal location is known as collection and transportation. The goal of proper disposal techniques is to reduce the negative effects on the environment and guarantee SW containment in order to avoid pollution. Recycling, which involves sorting and processing resources like glass, plastic, paper, and metals for reuse, is a crucial component of the disposal of SW since it lowers the need for fresh ingredients and lessens the environmental impact[2].

[1]Department of Mechatronics, School of Engineering, Kongu Engineering College, Perundurai, India. [2]Department of Civil Engineering, Sree Sakthi Engineering College, Karamadai, Coimbatore, India. [3]Department of Electrical and Electronics Engineering, Amrita School of Engineering, Amrita Vishwa Vidyapeetham, Bengaluru, India. [4]Center for Renewable on Microgrids(CROM), Huanjiang Laboratory, Zhuji, Shaoxing 311800, China. [5]Department of Electrical and Computer Engineering, Hawassa University, Hawassa 05, Ethiopia. [6]Department of Technical Sciences, Western Caspian University, Baku, Azerbaijan. [7]BROBOT Pvt. Ltd, Bengaluru, India. [8]Department of Electronics and Communication Engineering, I.T.S Engineering College, Greater Noida, Uttar Pradesh, India. ✉email: r_anand@blr.amrita.edu; baseem.khan04@gmail.com

By affecting ecological systems human wellness, acute and long-term effects, and the total preservation of our planet, improper SW management and disposal have drawn increased attention. Natural habitats are frequently destroyed as a result of the irresponsible dumping of SW. Ecosystem damage and deforestation can result from landfills, incineration plants, and mineral extraction for dump locations[3]. Numerous plant and animal species are displaced from their native habitats as a result of the loss of these habitats, which could result in extinctions or reductions in biodiversity. There are several ways that landfill waste contributes to a warming planet. CH_4, a powerful GHG that absorbs and retains far more heat than CO_2, is produced by landfills. Furthermore, significant carbon emissions are produced by the energy-intensive procedures used in waste manufacturing, transportation, and disposal[4]. The following table provides a succinct overview of the several sources of SW from the community, ranging from home waste to mining waste, which is easily illustrated in Table 1.

Innovations in the administration of garbage

The global level at which SW created is astounding. Over 2 billion metric tons of garbage were generated worldwide in 2016, estimates to the World Bank's data, and if current trends continue, predictions indicate that this amount will rise by 70% annually by 2050. Disparities in SW output are highlighted by the fact that high-income nations generate 34% of the world's waste although making up only 16% of its population. According to the thorough analysis, 3.40 billion tonnes of SW are anticipated to be generated worldwide in 2050. Overall, there is a positive correlation between SW generation and income level. Waste collection is a crucial stage in waste management[13]. Waste generation rates in that environment are very variable according on revenue[14]. In developing nations, only 48% of the garbage originates from metropolitan areas; however, this ratio sharply declines to 26% outside of urban regions.

International Garbage Content: Waste constituents vary by income level due to disparities in consumption habits. High-income nations produce about 32% of the world's garbage (minus food and green waste), while 51% of waste is made up of dry materials that can be recycled, Such as paper, cardboard, metal, glass, and plastic. Conversely, middle-income nations generate 53% of the world's SW[15]. Additionally, 20 percent of garbage is produced in low-income nations. Worldwide data from 2016 show that the processing and disposal of SW releases 1.6 billion tons of equivalent to CO2 greenhouse gases. Waste dumps as well as open dumps without gas collecting systems contribute to these high emission levels. Furthermore, if nothing is done in this area, it is predicted that pollution from solid garbage will account for 2.38 billion tonnes of carbon dioxide equivalents year by 2050. An unprecedented swelling population, and poor waste system oversight have made SW generation at the state level, as well as the national and international levels, a serious environmental concern[16]. Compared to other cities, major cities like Chennai, Coimbatore, and Madurai provide more to the total amount of waste generated.

AI-enabled machine learning techniques for Municipal SW Management (MSWM)

The global surge in urban population, economic expansion, and lifestyle changes has led to a critical escalation in MSW generation. Traditional approaches to waste management are no longer sufficient to deal with the growing complexity and scale of urban waste. This has prompted the integration of Artificial Intelligence (AI), particularly machine learning (ML) techniques, into the planning, optimization, and execution of MSWM systems. AI-enabled ML frameworks provide intelligent automation and real-time adaptability, offering a transformative shift from reactive waste handling to predictive and optimized resource management. AI-based MSWM lies the capability to model and learn from multi-source data, including socio-economic indicators, spatial zoning, household behavior, seasonal patterns, and historical waste records. Machine learning algorithms such as Support Vector Machine (SVM), Random Forest (RF), Naive Bayes (NB), and Artificial Neural Networks (ANNs) have been widely adopted for various prediction and classification tasks within MSWM. These models enable accurate forecasting of waste generation rates, spatial distribution of waste types, and emission potentials arising from landfilling or incineration processes. ML algorithms in waste management is their ability to handle non-linear relationships and high-dimensional datasets. For example, the Random Forest algorithm, known for its ensemble-based learning, performs well in classifying waste types based on numerous input features like population density, household income, and urban land use patterns. Similarly, Support Vector Machines provide high classification accuracy in distinguishing between different waste contributors when trained on

Waste Category	Sources
Quarrying Waste	Overload, scrap metal, and other waste products produced during extraction operations
Industrial SW	Waste produced by workplaces, retail stores, institutions, and businesses
Corporate Waste	Industrial byproducts, non-hazardous business garbage, construction and demolition debris, and manufacturing residues
Digital Waste[10]	eliminated electronics, including mobile devices, desktop computers, and gadgets
Renovation and Destruction Debris[11]	Construction and demolition debris, such as metals, bricks, wood, and concrete
SW	Paper, plastics, fabrics, glass, metals, food scraps, packaging materials, household waste, and yard waste
Healthcare Waste	Acupuncture needles syringes, and other biological waste are examples of infectious SW from healthcare institutions.
Urban Slurry	Sewage treatment facility residues, such as slime and biomass
Farm Waste[12]	Manure, insecticides, crop leftovers, and agricultural plastic SW
Unsafe Waste	Electronic garbage, batteries, chemicals, and solvents are among the wastes that are hazardous to the environment or human health.

Table 1. Types of Waste Involved with the Environment[5–9].

normalized and ranked datasets[17]. In recent studies, SVM has demonstrated Superior performance, achieving up to 96% accuracy in forecasting MSW volumes and associated environmental impacts, while maintaining computational efficiency suitable for real-time applications. Data preprocessing plays a critical role in enhancing the robustness of AI-based models. Techniques such as proximity ranking, feature extraction, dimensionality reduction, and outlier filtering help in isolating high-impact waste indicators. For instance, identifying plastic and organic wastes as dominant components in zonal contributions allows for more targeted prediction models and emission control strategies. These inputs, when paired with supervised ML models, offer an end-to-end solution from data acquisition to actionable insights for municipal authorities[18].

Beyond prediction, AI also supports optimization tasks within MSWM. Algorithms such as reinforcement learning and genetic algorithms are used for route optimization, bin placement strategies, and load balancing in waste collection systems. Moreover, unsupervised learning techniques like clustering and principal component analysis (PCA) facilitate the identification of waste generation hotspots, behavioral segmentation of residents, and operational inefficiencies across different urban zones. Deep Learning (DL)'s integration into smart sensor networks further strengthens MSWM systems. Real-time data from Internet of Things (IoT) devices such as smart bins equipped with fill-level sensors, gas detectors, and weight sensors can be streamed into AI models to dynamically adjust collection schedules, predict overflow risks, and detect illegal dumping[19]. By integrating AI with Geographic Information Systems (GIS) and temporal data analytics, waste management can be geo-visualized, enabling policy makers to implement localized and data-driven interventions. MSWM analysis using ML is the estimation of greenhouse gas (GHG) emissions and life cycle impacts of waste processing. ML models trained on emission profiles of biodegradable and non-biodegradable waste under various treatment pathways landfilling, composting, anaerobic digestion, and incineration can provide accurate carbon footprint assessments. This capability enables municipalities to simulate different policy scenarios and choose environmentally favorable treatment options. Reliable model performance depends heavily on the availability and quality of input data[20].

In many developing regions, historical records of waste generation, composition, and treatment are incomplete or inconsistent. Furthermore, the socio-economic diversity and informal waste sectors present additional modeling complexities. Addressing these issues requires building standardized data collection frameworks, integrating remote sensing technologies, and fostering public-private partnerships for data sharing. In Erode City, the application of ML algorithms model's ability to incorporate variables such as plastic accumulation, food waste content, population density, and income levels highlights the effectiveness of AI in supporting city-specific waste planning[21]. The deployment of proximity matrices and zonal clustering enhances the interpretability of the model, allowing local authorities to identify critical areas needing intervention. Further, AI-based MSWM lies in hybrid frameworks that combine deep learning models (such as CNNs and RNNs) with domain-specific heuristics and optimization layers. These can be employed to not only forecast waste volumes but also simulate long-term sustainability outcomes. The integration of explainable AI (XAI) can further assist municipal decision-makers by providing transparent rationale behind model predictions, improving trust and adoption[5]. High-income countries, despite comprising only a fraction of the world population, contribute disproportionately to waste generation. Furthermore, landfill sites without adequate gas management infrastructure significantly contribute to greenhouse gas emissions, reinforcing the critical link between waste management and climate change mitigation. In response to these challenges, recent advancements have seen the integration of AI, particularly ML techniques, into solid waste management systems. These technologies offer promising avenues for forecasting waste generation, optimizing collection routes, classifying waste types, and quantifying emissions. Machine learning models such as SVMs, RFs, and NB have demonstrated effectiveness in handling complex, nonlinear datasets derived from socioeconomic indicators, spatial zoning, and historical waste data. Moreover, real-time data acquisition through IoT-enabled sensors and deep learning methods has the potential to revolutionize waste segregation and operational efficiency.

This study aims to develop a robust, data-driven framework utilizing machine learning algorithms for accurately forecasting municipal solid waste generation and emission dynamics in Erode City, India. The primary objectives are to (i) integrate diverse socio-economic, spatial, and historical waste data to train predictive models, (ii) identify key waste components and influential factors through feature selection and proximity analysis, and (iii) evaluate the performance of multiple supervised learning techniques to select the optimal model for real-time waste management applications. In addition, the research seeks to provide actionable insights for policymakers by quantifying carbon emission potentials associated with different waste treatment scenarios, ultimately guiding sustainable waste handling strategies and infrastructure planning.

Socio-economic impacts due to the unregulated open dumping of SW

Sanitary rubbish dumps are carefully planned structures intended to reduce environmental effect and safeguard public health, whereas open dumping is an uncontrolled and environmentally damaging manner of disposing of waste. Their design, regulations, long-term sustainability, and environmental effects are where the main distinctions may be found. Also known as unsecured dumps or uncontrolled SW, non-engineered waste dumps can have a significant adverse environmental impact due to poor planning, construction, and management. Notably, these impacts vary depending on climate, waste mix, and local regulations[6]. The ones that follow are some common environmental impacts associated with poorly designed landfills:

Freshwater Pollution: One of the main issues with open dumps is that there is no protective cover or leachate collection equipment, which allows dangerous materials to seep into the groundwater[7]. The quality of groundwater can be seriously threatened by leachate, a liquid that is created when water flows through garbage and can contain contaminants such organic compounds, poisons, and heavy metals.

Rainfall can carry pollutants from the dump into neighboring waterways, wetlands, or streams, resulting in surface water contamination as well as air pollution. The marine environment and water quality may be impacted by contaminants such as suspended particles, nutrients, and dangerous compounds that are carried by this discharge[8]. Massive amounts of harmful gases are produced during natural decomposition processes, which typically lead to emissions of CH4, NOX, COX, and SOX. These emissions significantly contribute to the phenomenon of climate change. Toxic gasses, volatile organic compounds, and particulate and gaseous debris are emitted as a result of open burning operations in non-engineered landfills[9]. Consistent with the aforementioned, Fig. 1 illustrates how pollutants impact respiratory health more broadly and contribute to the deterioration of air quality.

Deterioration of Ground and loss of Habitation: Poor waste management can lead to the depletion of productive topsoil and erosion of the soil. Non-biodegradable materials building up in the soil can hinder plant development and upset regional ecosystems. Natural habitats are frequently invaded by landfills, which causes plants and animals to be displaced[22]. Changes to local ecosystems have the potential to upset the ecological equilibrium and impact upon the existence of different species

Hazards to the General Population and Surroundings: Non-engineered landfills may be unattractive and have a detrimental effect on the surrounding area's aesthetics. In addition to lowering property values and discouraging tourists, this aesthetic blight can also lower the standard of living for locals. Rodents, bugs, and insects are among the pests drawn by vacant dumps, and they can transmit diseases to neighboring communities[23]. People who live close to non-engineered landfills may be at considerable risk for health problems due to exposure to hazardous materials in the soil, air, and water.

Long-lasting Ecological Tradition: Even after being formally closed, non-engineered landfills may continue to discharge pollutants and present environmental hazards for many years. Proper waste management techniques, such as recycling, SW minimization, and the construction of engineered landfills with suitable containment and treatment systems, are necessary to address the environmental effects of non-engineered landfills[24]. Furthermore, in order to lessen these effects and encourage sustainable waste management, public awareness campaigns and legislative actions are essential.

The importance of reporting systems for garbage dumps

ccording to Su et al. (2024), synthetic chemicals are present everywhere as a result of growing industry, commercialization, and waste disposal methods. In developing nations like India, a large number of landfills lack cover materials and are not engineered, which results in a higher level of contaminant emissions inside the designated region (Cheu et al. 2025). According to the explanation above, SW dumps need more care because they release more infectious gases in a shorter amount of time (Dong et al., 2025). Over time, several kinds of gasses will emerge from the landfills. Huge amounts of important gases, including CH4, H2S, NO2, CO2, and CO, are discovered (Tamiru et al. 2024). GHG emissions from SW disposal sites account for about 2.8% of all worldwide emissions (Ali et al., 2024). For the last twenty to thirty years, the presence of GHGs in an area in particular has become of significant importance due to its dynamic character (Min et al. 2024).

According to a 2022 International Energy Agency assessment, methane emissions are responsible for a 30% increase in global temperatures. 3R principles, awareness campaigns, and creative techniques that prevent adverse environmental effects will be taken into consideration in order to develop improved SW handling strategies (Wang et al. 2025). Recycling techniques will eventually result in a significantly lower amount of waste being delivered to disposal yards, which will provide an atmosphere without pollution (Wang & Sun et al.

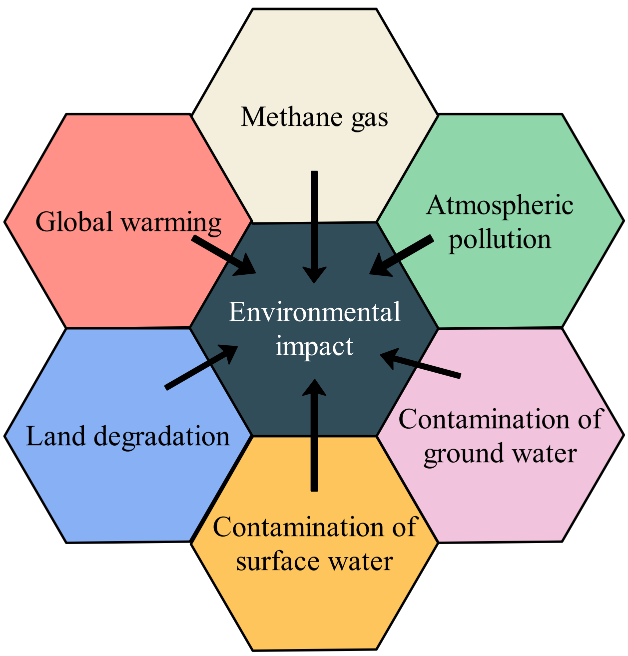

Fig. 1. SW impact on the environment.

2025). However, because of the lack of resources and exposure to technology, this approach takes time to fully adopt in developing nations[9]. These unplanned disposal sites are still present in many Indian towns, causing harmful emissions, unpleasant odors, and other environmental problems at the abandoned site (Li et al. 2021). In the current situation, it is becoming more difficult to characterize waste according to its differing significance, identify greenhouse inventories, analyze the likelihood of energy recovery possibilities, and implement cost-effective thermal treatment procedures.

Successful disposal of SW is proving to be extremely difficult for the sanitary department, mostly because of a lack of real-time data that makes it difficult to make precise predictions for the future. In accordance with current information collection and retrieve facilities, electronic sensors will be utilized to continuously monitor the infectious discharges from landfill sites (Dawar et al. 2024). Simulations to forecast greenhouse gas emissions from economically diverse countries are also conducted using the emission data sets (Vyas et al. 2022). By combining the SD Framework with the IPCC platform, additional waste prediction approaches can help with the appropriate summarization of the discharge statistics from a particular area for a certain period. In order to depict the gases released from urban garbage processing facilities, the majority of the literature concentrated on empirical modules, first order decay techniques, and GHG indicators (Xu et al. 2021). Conducting a behavior analysis of the waste characteristics is still a challenging task because SW is fluid in nature and depends on environmental conditions, zone types, processing facilities, and external stresses[25–27]. An internal technology for tracking must be suggested in order to accurately forecast the emissions from certain landfills. This will help decision makers suggest improved SWM techniques. SW sorting, pickup path layout, garbage creation, immediate form dump emptying rates, illegal dumping prediction, and operational issue identification can all be projected to be valuable by integrating field research with learning-based approaches.

Machine learning & deep learning for real-time waste sorting and recovery

Arun et al. (2025) introduced the WMRDL framework, an integrated system combining IoT-enabled sensors with advanced deep learning classifiers to streamline real-time waste sorting and material recovery. The IoT sensors continuously transmit data from facility-level sorting units to a deep learning model that identifies recyclable components. As a result, the system enhanced recovery efficiency by around 30%, reduced manual labour, and significantly lowered operational costs demonstrating strong alignment with circular economy. Meanwhile, a related approach detailed by Mosharof Hossain Dipo et al. (2025) implemented a YOLOv12based object detection model for live waste identification in urban settings. Achieving both speed and precision, the model enabled robust classification even in cluttered environments, supporting automated decision-making in smart-bin deployments.

In early 2025, Qiu, Xie, and Huang (2025) proposed an improved EfficientNetV2 architecture featuring a ChannelEfficient Attention (CEAttention) module and lightweight multi-scale feature extraction (SAFM). This model enhanced critical feature retention while maintaining computational efficiency, achieving 95.4% classification accuracy on real-world waste datasets demonstrating its Suitability for edge devices and IoT-integrated sorting stations. Complementing this, a 2024 study by Pillai et al. reviewed the widespread application of DenseNet121 via transfer learning in waste classification pipelines. These models achieved classification accuracies ranging between 91% and 96.4%, particularly when combined with genetic-algorithmbased parameter tuning and GradCAM visuals for interpretability. From a systems optimization angle, recent literature (Ahmed et al., 2025) integrates AI with IoT, genetic algorithms, and evolutionary optimization to improve waste segregation, collection scheduling, and waste-to-energy strategies. The review emphasizes real-time monitoring through smart bins, automated decision-making, and the synergy with circular economy goals. Additionally, AI-enhanced thermal treatment processes (including pyrolysis and incineration) have been optimized through control algorithms to maximize energy recovery and minimize emissions. Table 2 elaborates the exploration of ML and DL techniques in the waste management platforms.

Recent studies have emphasized the significant role of artificial intelligence (AI) and machine learning (ML) in optimizing MSWM systems. Lee et al. (2024) proposed an IoT-integrated waste collection framework in smart cities, utilizing sensor-based smart bins and XGBoost-based forecasting models. Their system dynamically adjusted collection routes in response to bin fill levels and traffic congestion, resulting in a 32% reduction in fuel consumption and improved route efficiency. Similarly, Mustafa and Zhang (2025) developed an advanced image-based waste classification system using a ResNet-50 deep learning backbone and optimized it with Artificial Ecosystem Optimization (AEO). This approach significantly improved automated waste sorting accuracy, achieving over 95% precision across mixed waste categories, thus reducing manual handling in material recovery facilities. In parallel, Patel and Singh (2025) introduced a Green AI framework that focused on energy-efficient waste management by integrating lifecycle assessment tools with lightweight ML models. Their approach minimized computational load while enhancing classification accuracy and reducing carbon emissions from collection and transport systems. These advancements demonstrate a clear trend toward embedding AI not only for operational efficiency but also to align MSWM strategies with sustainability and circular economy goals.

Zonal site mapping and socioeconomic survey analysis of SW generation and disposal

Continuous monitoring and investigation of garbage management and its monetary viability rate are the focus of the current study project in Erode City. In the Tamil Nadu environment, Erode City is Listed as the fifth largest urbanization. The entire area is 5722 square kilometers, of which 5031.36 square kilometers are sprawling cities and the balance of 690.64 square kilometers are farmland. According to the Parametric Progression approach, the city would have 25.21 lakh residents by 2022. The elevation of Erode City is 171.91 meters above mean sea level. A business entity, two administrative sections, four councils, three hundred and seventy-five commercial small towns, two hundred and twenty-five community councils, ten taluks, and fourteen improvement zones

Criteria	Machine Learning (ML)	Deep Learning (DL)
Model Types	Decision Trees, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN)	CNNs (Convolutional Neural Networks), RNNs, LSTMs, YOLOv5/v8, EfficientNet, DenseNet
Input Requirements	Structured/tabular datasets (numeric or categorical data)	Large-scale image, video, and sensor data (e.g., waste images, real-time bin camera feeds)
Accuracy Range	Typically, 75–90% depending on features and preprocessing	Generally, 85–98% for image-based classification tasks when trained on sufficient data
Training Time	Faster training and lower computational cost	Requires longer training and GPU support for convergence
Interpretability	High interpretability using decision boundaries, feature importance, or SHAP values	Low interpretability; often considered "black-box" models, though Grad-CAM or attention maps help visualize model focus
Use Case Examples	- Predicting waste generation trends- Classifying landfill site performance- Routing optimization	- Automated waste sorting in smart bins- Real-time waste object recognition- Semantic segmentation for garbage piles
Advantages	- Simpler to implement- Works well on small datasets- Requires less hardware	- Captures complex patterns in high-dimensional data - Better generalization for visual and spatial tasks
Drawbacks	- Poor performance on unstructured data (images/videos)- Needs manual feature engineering	- Needs large labeled datasets- High computational demands- Slower retraining with evolving data
Best Fit Scenarios	Tabular data from municipal databases, IoT sensor readings, questionnaire responses	Smart waste monitoring systems using cameras, UAV footage, or robotic waste sorting setups
Recent Trends (2024–2025)	- Integration with GIS and IoT- Hybrid ML+heuristic optimization for routing and collection frequency prediction	- Lightweight DL models for edge deployment (e.g., MobileNet, TinyYOLO) - Vision transformers and attention-based DL for waste image segmentation

Table 2. Comparative statistics of ML and DL application in the waste management platform[28–36].

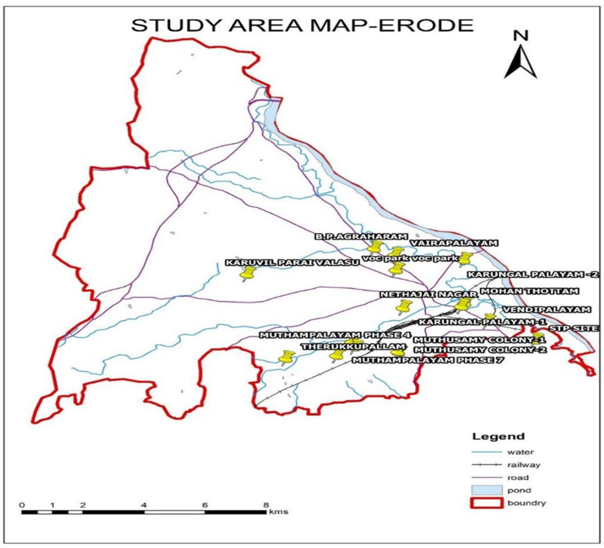

Fig. 2. Site identified for research study on examining SW.

constitute the city. The city generates an average of 250 metric tons of waste daily, with dumping sites located outside the city Limits. In the case of Erode, for the past 61 years, SW has been disposed of at the Vendipalayam yard, which spans 19.45 acres, and the Vairapalayam yard, covering 7.4 acres as shown in Fig. 2.

The increasing particulate and gaseous emissions at both dumping yards have led to frequent fire incidents, foul odors, groundwater contamination, and health hazards like carcinogenic diseases, which have sparked periodic protests from residents living nearby. Primary data related to SW collection and management practices were obtained from the Erode city municipal corporation. The business headquarters also provided secondary data, such as the variety and size of collection trucks, the assortment of storage bins, the types and characteristics of garbage, the locations of disposal sites, the daily rates for debris generation, and information on the collection system. From residence-to-residence garbage pickup has been a popular practice in the region as the city develops within the agenda of the Smart City Mission (Darmian et al. 2020). Following collecting, garbage is taken to the disposal grounds for additional destruction; at the Vendipalayam location, this process results in considerable atmospheric emissions. The complete methodology of the proposed work is illustrated in Fig. 3.

Questionnaire Survey: A clear series of written inquiries with multiple choice answers created for an opinion poll or empirical research is called a survey. It helps to document how a certain set of people performs in a given field. In the framework of garbage disposal for Erode City, the survey's findings shed light on the requirements

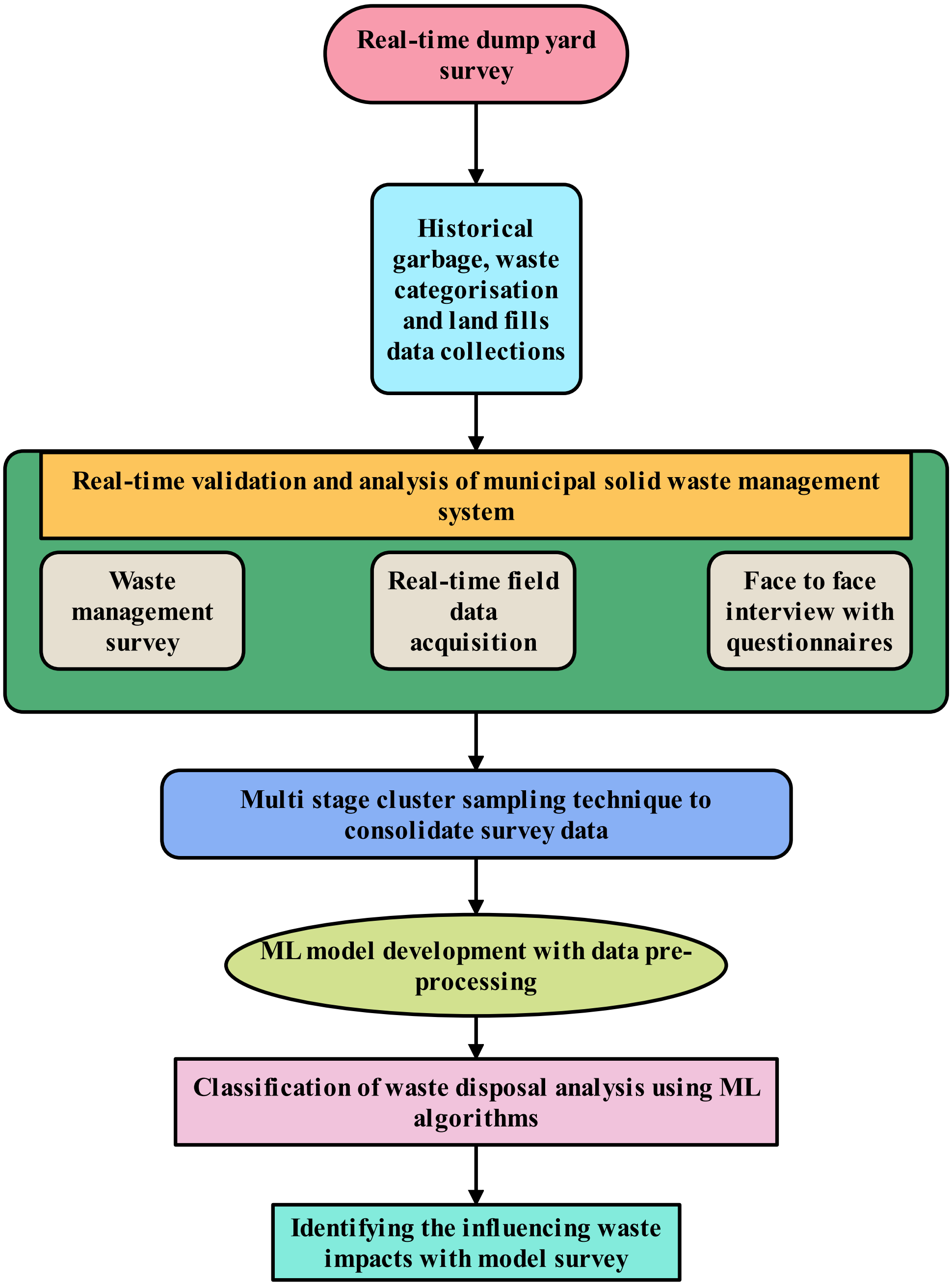

Fig. 3. The overall methodology of the SW disposal and management of the proposed work.

and Suggestions of the respondents. A waste handling Survey is conducted using a scientific methodology and involves in-person interviews. For eight months, from September 2023 to April 2024, the waste management status of households is surveyed using the Multi Stage Cluster sample technique, taking into account both summer and winter conditions.

The number of participants needed to carry out an investigation in the research region is calculated using Yamane's (1967) method (Equation (1)), where N is the total metropolitan populace, e is a level of accuracy 0.07, and n is the necessary sample size.

$$n = \frac{N}{1 + N(e^2)} \tag{1}$$

As previously stated, around 25.21 lakh people are expected to live in the city limits based on a scientific forecast approach. The appropriate sample size is calculated using the aforementioned computation (Equation (2)) (Levis et al. 2013).

$$n = \frac{25,21,953}{1 + 25,21,953(0.07^2)} = 204 \tag{2}$$

Surveys are distributed to people in various locations, including institutions, industries, residences, shops, and apartments, based on zoning and in accordance with directives from corporation officials. Separation was then carried out in order to meet the needs, forecast the reactions, and enhance the processes for handling SW. The waste is collected from specified locations using 450 dumper bins and more than 500 collecting trucks. Socioeconomic characteristics have a significant impact on the amount, kind, and frequency of waste produced by communities, according to the suggested solid SW prediction. These factors represent human behavioral, economic, and demographic conditions that directly or indirectly affect consumption patterns and, consequently, waste disposal rates. Based on the literature and standard practices in environmental modeling, the following socio-economic indicators are typically incorporated:

- **Population Size and Density** - The total population and population density are fundamental indicators, as larger populations inherently produce more waste. Population density can influence the types of waste (e.g., more packaging in urban areas) and collection logistics. Data for this variable is usually obtained from national census data or municipal demographic reports.
- **Household Income or GDP per Capita** - Average household income or regional GDP per capita is a critical driver of waste generation. Higher income levels are associated with increased consumption of packaged goods, food waste, and disposable items. Studies (e.g., World Bank, UNEP) demonstrate an advantageous connection between individual garbage generation and revenue. This data is typically sourced from national economic surveys, statistical yearbooks, or international datasets such as those from the World Bank or IMF.
- **Education Level** - The average educational attainment of a population often reflects awareness of sustainable consumption and waste segregation practices. While higher education levels may lead to more waste generation due to affluence, they also correlate with better recycling behaviors. Educational data is obtained from census bureaus or education departments.
- **Urbanization Rate** - The percentage of the population residing in urban areas reflects the density of service demands and types of waste. Urban settings tend to generate more commercial and packaging waste compared to rural areas. Urbanization statistics are provided by national planning authorities and international databases like the UN World Urbanization Prospects.
- **Employment Rate and Economic Activity** - The structure of employment particularly the distribution across industrial, commercial, and service sectors can influence waste types (e.g., more industrial or office waste). Employment and economic activity data can be gathered from labor departments or economic development boards.

Proposed methodology

The methodology adopted in this study encompasses a multi-layered experimental framework integrating real-time data acquisition, systematic waste characterization, and machine learning-based classification models to optimize MSWM systems. The initial phase involved conducting an extensive real-time dump yard survey across selected urban zones, where waste generation and disposal practices were closely monitored. Data was collected directly from landfills and disposal sites, focusing on volume, type, and source of waste materials. Geo-tagging was employed to record the spatial distribution of waste hotspots, while time-stamping allowed for temporal pattern analysis of waste generation. To complement the real-time data, historical datasets on garbage composition, landfill utilization, and municipal waste handling records were retrieved from public domain sources and municipal archives[28]. These records were digitized, categorized, and synchronized with the field survey data to ensure longitudinal coherence. The survey methodology was structured using a face-to-face interview format supported by detailed questionnaires targeting key stakeholders, including municipal staff, sanitation workers, and local residents. A multistage cluster sampling approach was used to cover heterogeneous urban segments ranging from densely populated slums to planned residential areas to ensure that the sample accurately reflected socio-economic and geographical diversity. The complete methodology of the proposed work is illustrated in Fig. 3.

Real-Time Data Collection and Survey Design: A real-time dump yard survey was conducted across identified municipal zones, targeting active and legacy landfill sites. The field data acquisition involved:

- On-site waste classification (organic, plastic, e-waste, inert materials, etc.),
- Volume estimation and source tracking, and
- Geo-tagging and temporal tagging of disposal patterns.

Additionally, historical data on waste categorization, landfill usage patterns, and municipal records were digitized for comparative analysis.

Survey Execution and Sampling: The primary data was collected using face-to-face interviews and structured questionnaires tailored to municipal staff, waste workers, and households. A multistage cluster sampling technique was employed to ensure representative coverage across zones with varied population densities and waste profiles [28]. This method minimized sampling bias while ensuring statistically significant results.

Data Processing and Model Development: The consolidated data underwent several pre-processing steps:

- Data normalization and encoding to manage heterogeneity,
- Noise reduction and outlier elimination using interquartile range analysis, and
- Missing value imputation via k-nearest neighbor (k-NN) interpolation.

After preprocessing, supervised machine learning algorithms were applied for classification of waste disposal patterns. Algorithms such as Random Forest, Decision Trees, and Support Vector Machines (SVM) were evaluated based on accuracy, precision, recall, and F1-score metrics.

Model Evaluation and Impact Assessment: The trained models were validated using k-fold cross-validation (k=10) to ensure generalization. The model with the highest classification accuracy was employed to:

- Identify key influencing factors affecting improper disposal (e.g., lack of awareness, logistics, zone density), and
- Predict waste generation trends and hotspot zones for future policy planning.

The most effective model identified through 10-fold cross-validation was deployed to classify various waste disposal behaviors and to assess the impact of influencing variables such as locality type, income level, frequency of collection, and availability of segregation facilities. The model further assisted in predicting future waste accumulation trends and identifying critical zones prone to mismanagement. The entire workflow, including data acquisition, preprocessing, model training, and validation, is depicted in the Fig. 3, which illustrates the methodological pipeline used to establish a data-driven decision support system for efficient SW governance[30].

Real time experimental result interpretations and discussion

In the proposed model, the socio-economic factors used to predict SW generation include population density, average household income, urbanization rate, education level, employment rate, and per capita food expenditure. These variables are commonly associated with consumer behavior and waste generation patterns, particularly in urban contexts[31–33]. Surveys are distributed to people in different places, including educational institutions, companies, residences, stores, and housing, based on zoning and in accordance with directives from municipal authorities as represented in Fig. 4. Isolation was then carried out in order to meet the needs, forecast the reactions, and enhance the processes for handling SW. The waste is collected from specified locations using 450 dumper bins and more than 500 collecting trucks.

For model training, historical data for all selected socio-economic indicators were collected for the target region over a span of 5 years (2019-2023). Sources included:

- National statistical agencies
- Local municipal records
- Household Income and Expenditure Surveys (HIES) through Questionnaires

These datasets were pre-processed, normalized or standardized and missing values were addressed using imputation methods such as mean substitution or regression imputation. To prepare the algorithmic learning models, they were then combined with relevant SW production statistics (in tonnes). The identical demographic and economic data were projected for future periods in order to make predictions. These were sourced from the Government planning and forecasting reports. The reliability and completeness of socio-economic data are critical to the accuracy of waste prediction models. Hence, only validated sources with temporal and spatial consistency were used[34–36].

Figure 5 makes it quite clear that, aside from residences and inhabitants, no zone has good access to the door-to-door collection method. The resulting rubbish is therefore exceedingly difficult for people to dispose of in communal and waste dumpsters. Consumers find communal bins to be easier to use than other options like free waste disposal and solidarity rubbish receptacles. Therefore, disposal won't pose a concern in the research area provided the gathering infrastructure is enhanced. According to the analysis of the replies, rubbish gets collected once every two days on average. Waste is gathered every day on a quarter of the municipality's property, which expands the potential for an even better collecting network.

The waste partition quality of individuals at the point of origin is depicted in Fig. 6, which reflects the current situation's absence of knowledge regarding garbage disposal techniques. On the one hand, those surveyed who live in condominiums and are cognizant of the source segregation phenomena are able to absorb it. Conversely, those with an industrial background, as well as those who work in stores and institutions in the study area, are

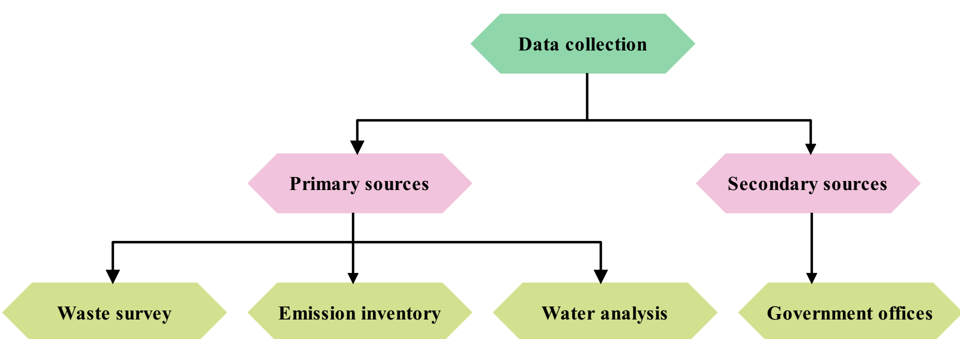

Fig. 4. Data survey undertaken resources and its attributes.

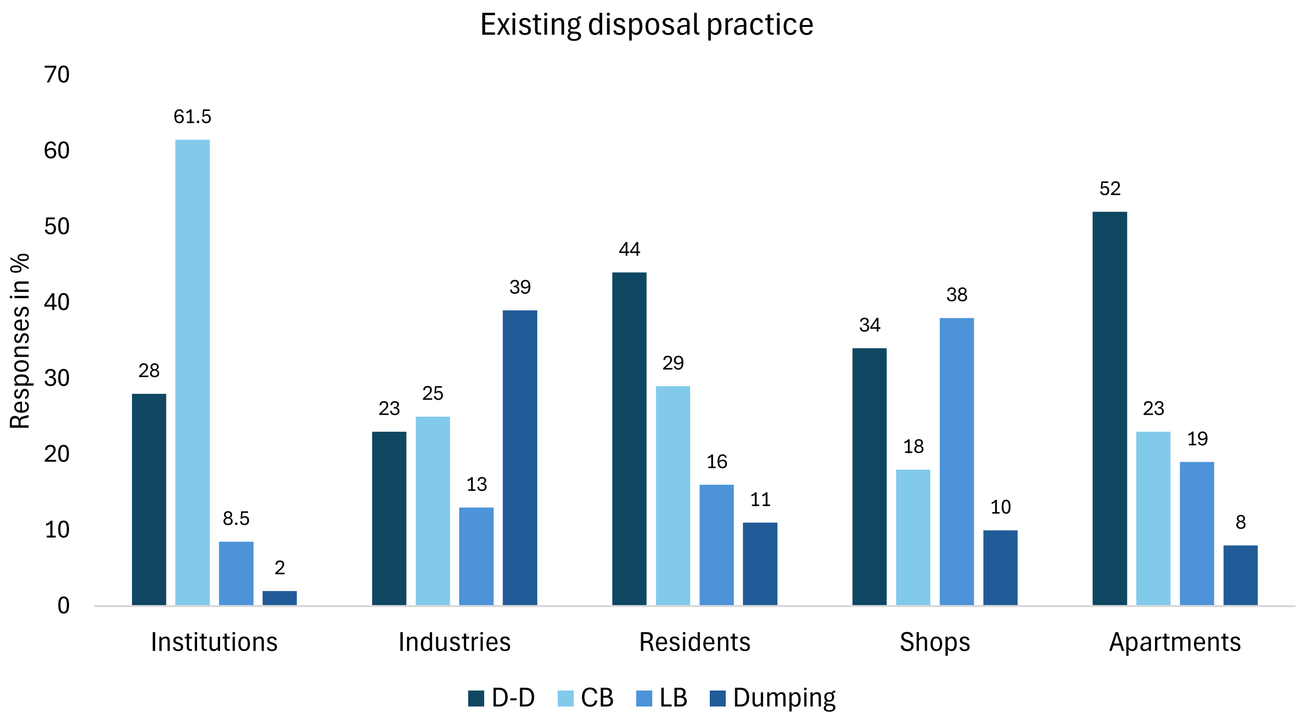

Fig. 5. Existing disposal techniques inferred from real-time field survey.

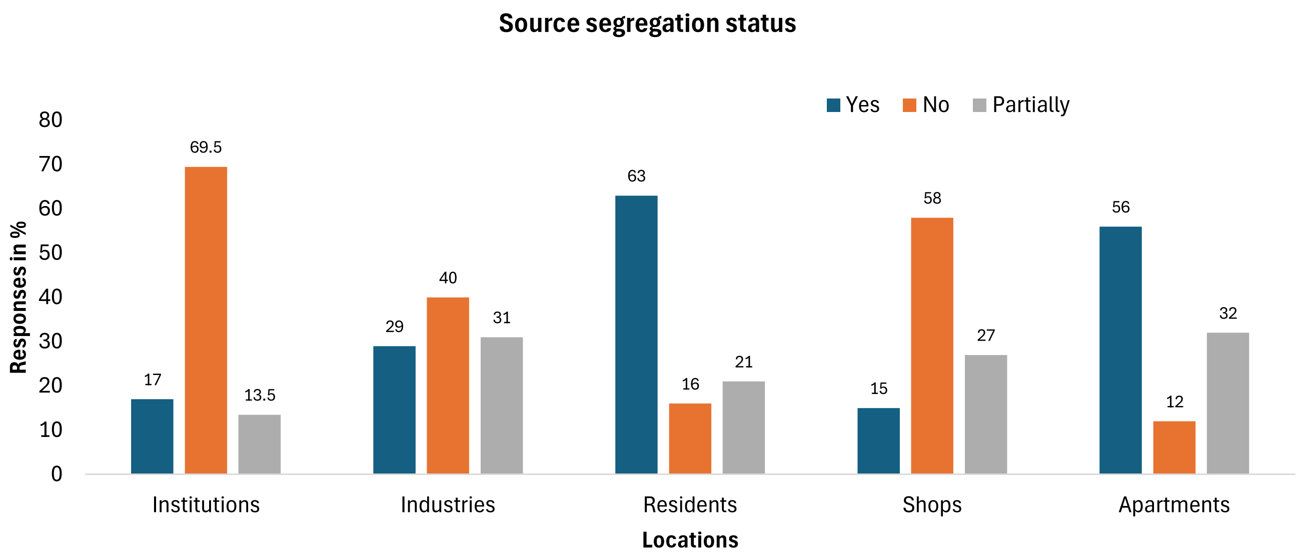

Fig. 6. Source segregation status received from field survey.

less knowledgeable about source segregation techniques. Therefore, in order to guarantee superior separation aspects, company authorities might enable recycling initiatives, combined initiatives, and other actions for them

According to the waste disposal study, the main makeup of waste is determined by several zone characteristics, which are shown in Fig. 7. Accordingly, a range of replies were recorded by the respondents. For example, respondents from businesses, establishments, and industries selected paper as the primary waste material, whereas respondents from residential areas, such as flats, indicated food waste as the primary waste material. Therefore, it may be deduced that the responses were evenly recorded according to the zone to which they belong. Additionally, it can be deduced that survey participants advocate for a variety of disposal techniques. Burning, landfills, incineration, and composting are among the techniques mentioned. Remarkably, the majority of respondents were knowledgeable enough to select composting as the suggested method of disposing of waste, which makes sense to use.

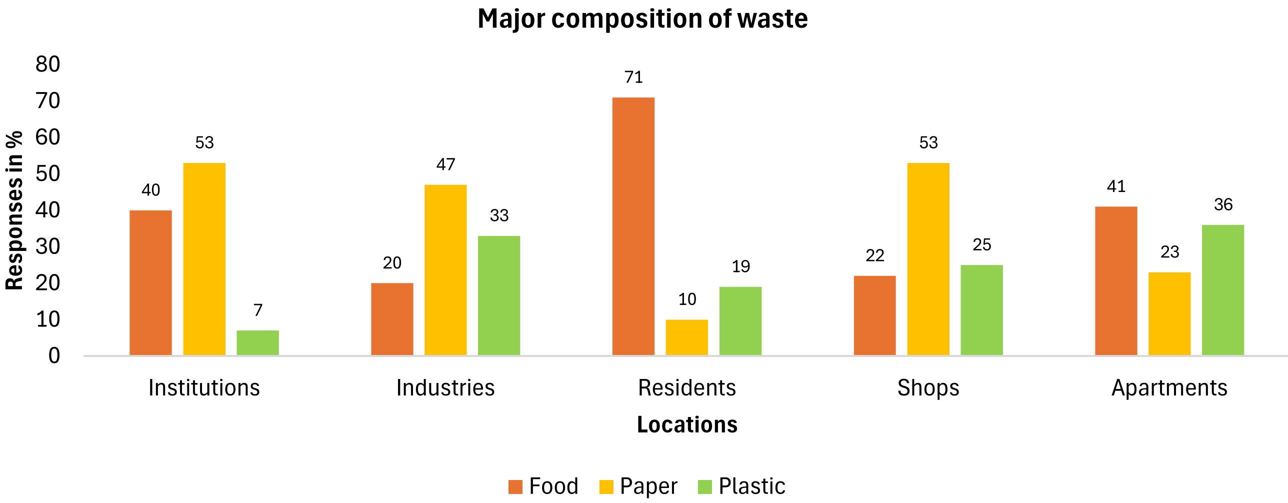

Fig. 7. Major composition of waste followed in erode city.

The respondents have documented their main suggestions for managing and enhancing the gathering services in their neighborhood. Strategies including offering incentives for recycling, developing new or clever technologies, enacting strict legislation, and launching awareness campaigns are all recommended. Among the environmental problems that are now plaguing the city, waste pollution is of more concern to the citizens than air and water pollution. Improved collection and disposal methods should be created and put into place for greater economic stability in light of the field visits and questionnaire replies. The majority of interviewees' replies demonstrate their concern about the methods used for collecting and disposing of waste in the research area. Additionally, Erode City's citizens are willing to pay for garbage pickup services in accordance with their sincerity. It is anticipated that the amount of waste generated will decrease if the cost of collection is directly linked to the organisation generating the waste. Pricing for waste collection can be determined by factors such as the quantity of waste collected, the frequency of collections, the weight of the waste, and the quantity of bags created (a practice known as "bag-and-tag pricing"; Metin et al. 2003). SW output will progressively decline if price is determined by the amount of waste generated by the household. Additionally, experimental research demonstrated that unit-based waste price improves recycling efficiency and dramatically reduces household waste creation.

Results and discussion

Numerous socioeconomic aspects must be considered in order to accurately forecast the pace of SW generation and emissions. Typically, SW quantities are considered together with socioeconomic and demographic aspects while developing SW forecast models. This study employs efficient learning techniques, Such as Random Forest, Support Vector Machine, and Nave Bayas, to forecast the rates of SW generation. First, the data is pre-processed using Data Statistics program 4.3 computer program to provide effective datasets that support high-quality models[22]. With the exception of garbage and silt, the municipality in the current study region is not proclaimed to collect all waste streams. After regular data changes using the IDE platform, the most definite dataset has been taken into consideration, along with precise waste volumes.

When data is lacking or extremely impracticable, or when the entire amount of waste in tonnes is missing, data reconciliation encourages the substitution of standard deviations for historical records. The amount of waste that is disposed of and recycled is taken into account when calculating the overall amount of MSW. The analysis uses the total amount of waste that ends up in landfills, which is gathered from company representatives and shown in Table 3. Without any data adjustments, this is the initial information that is then added as a whole dataset.

The computerised learning techniques' settings have been continuously changed to lower testing and training errors in order to assess the prediction process's feasibility and optimistic behaviour. Following the algorithm's listing of the best-fitting model, both sets of data are chosen at random to continue training and testing the model. By conducting the analysis as shown in Fig. 8, indices such as the mean and standard deviation for each algorithm are computed in order to clarify the performance. The overall parameters taken into account for the prediction model are listed in Table 4 according to their importance in the SW survey. The most common characteristics influencing waste generation, according to an assessment of models on SW generation projections, are household size overall, income level, and educational attainment[23]. Official historical records of the factors (qualitative and quantitative) influencing the development of SW are lacking, particularly in emerging nations. Since there are variations in the degree of correlation between each of these traits. Therefore, a prediction in one level does not always translate to another level. These represent the primary constraints on the estimation of MSW production. The published models exhibit diversity in their application, ranging from counties to households. The waste stream selection is critical to successful modeling. Regression analysis and correlation formed the foundation of the majority of the models. It is evident that total garbage and silt quantities

ANNUAL REPORT (MT/Day)										
	2019		2020		2021		2023		2024	
Month	Silt	Garbage	Silt	Garbage	Silt	Garbage	Silt	Garbage	Silt	Garbage
JANUARY	791.4	2566.65	283.92	6728.3	164.64	5348.98	452.55	3731.22	532.49	3076.55
FEBRUARY	814.2	3699.74	1464.05	5695.9	410.1	3928.96	500.17	3324.19	560.1	3249.76
MARCH	807.4	4774.51	569.57	5121.63	78.34	5436.5	18.75	3848.55	567.58	3724.18
APRIL	297.3	383.41	220	5048.53	74.78	4923.5	105.5	3086.78	392.91	2810.64
MAY	364.3	402.64	324.1	5024.5	10.9	3465.8	505.9	4667.45	422.82	3473.73
JUNE	409.45	736.22	383.44	5458.8	42.87	4868.2	260.63	3556.76	364.27	2576.22
JULY	432.59	437.32	332	5175.08	30.64	5133.87	376.89	3610.73	300.22	3177.32
AUGUST	286.13	283.96	291.76	5594.16	1326.82	5583.82	452.66	3932.14	460.89	2787.62
SEPTEMBER	172.60	828.44	149.4	6420.5	149.41	6420.87	459.39	3950.25	626.05	2716.91
OCTOBER	68.48	835.13	268.4	6457.28	268.39	6457.28	154.73	3740.93	322.37	2856.81
NOVEMBER	242.4	1077.6	71.49	5489.44	71.49	5489.44	3791.4	3069.06	300.87	2407.04
DECEMBER	340.6	1228.81	41.99	2392.96	104.06	5519.95	450.58	3233.39	341.08	2797.28

Table 3. Rate of Waste Elimination for a Five-year Period (2019-2024).

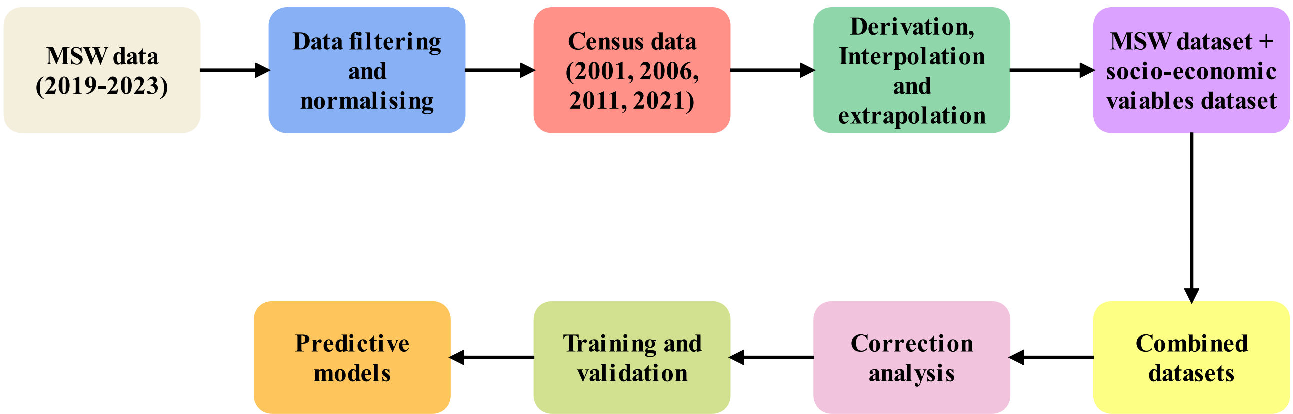

Fig. 8. Prediction model for waste management.

Analyzing Data	Standards for Inspection	Variable Forecast	The ranking/weighting factor
Total Assessment of Waste Solids	Wood and Linen produced for each zone	influenced the inquiry's communal setting criterion	6895
Assessment of Waste from Plastics	Obtained Plastic Scraps	influenced the inquiry's communal setting criterion	12992
Culinary Dump	Obtained Biological dumps	influenced the inquiry's communal setting criterion	10376

Table 4. Garbage Estimation Framework Inputs.

are growing higher based on the Five-year dataset based on the ranking, pre-processing operation is executed to obtain precise range of waste generation and disposal quantities.

Inadequate waste handling: The growth in garbage quantities could also indicate issues with waste management systems, such as insufficient waste collection, poor recycling efforts, or illegal dumping practices. The increase in total garbage suggests that either new sources of waste have been added or that older systems are not equipped to manage growing waste production.

Improper drainage and erosion: If drainage systems are overwhelmed, they could also contribute to the accumulation of silt and garbage in water bodies.

Higher silt in specific seasons: Periods of higher rainfall (or flooding) can increase soil erosion from agricultural fields and construction sites. This could explain higher silt levels in certain months. Similarly, the garbage entering water bodies could increase due to rain washing debris from urban areas into rivers.

Infrastructure development, such as roads, buildings, and dams, is often associated with increased siltation. Without adequate sediment control measures (such as barriers or silt traps), loose soil from construction sites can easily enter water bodies, increasing silt loads. From Fig. 9, it is concluded that the rise in total garbage and silt quantities likely stems from a combination of urbanization, inadequate waste management systems, deforestation, and increased construction activities, possibly aggravated by seasonal and climatic factors.

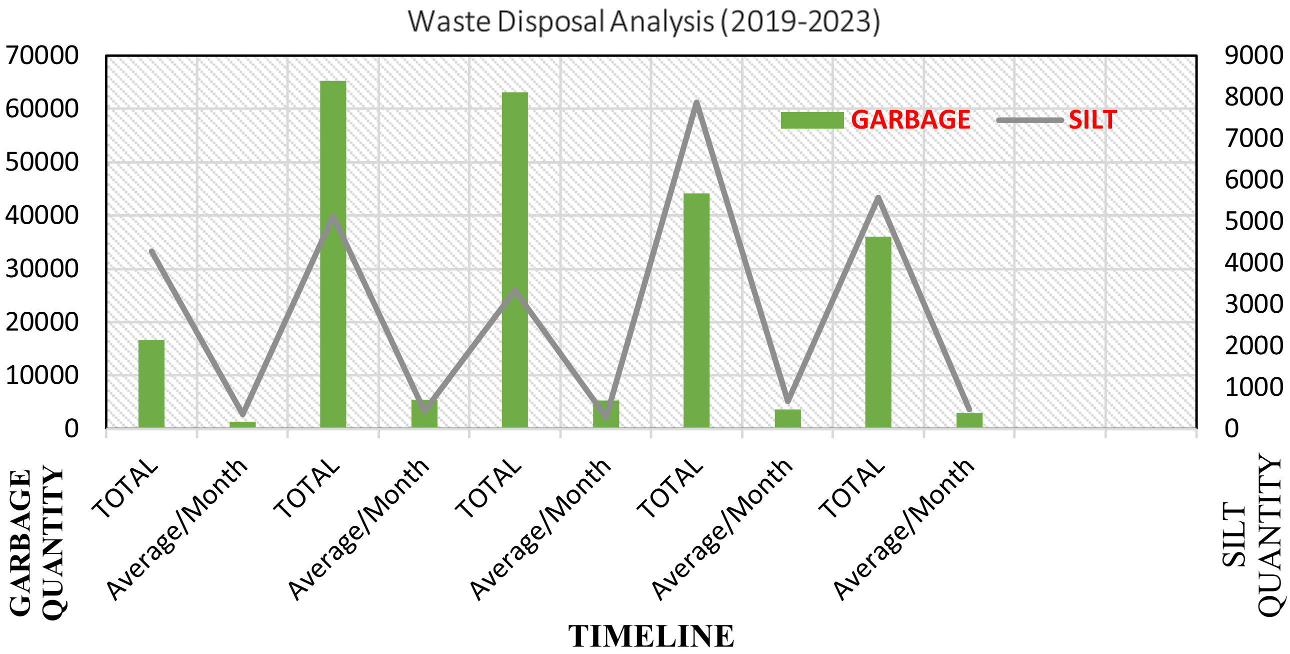

Fig. 9. Waste disposal analysis.

Data modeling on soild waste prediction using machine learning algorithms on soild waste types availability

A crucial component of sustainable development is waste management, which calls for precise forecasting to maximise the distribution of resources and adverse ecological effects (Yong et al. 2019). Once the SW generation estimates are obtained, the next crucial step is to assess the carbon emission potential associated with the handling, treatment, and disposal of that waste. Carbon emissions from MSW primarily arise due to the degradation of organic matter in landfills, waste transportation, incineration, and various treatment processes. The estimation process generally follows standardized guidelines developed by international bodies such as the Intergovernmental Panel on Climate Change (IPCC) and other national waste inventory frameworks. Carbon emission estimation starts with a detailed waste composition analysis. MSW typically contains various components such as:

- Biodegradable organic SW, such as food scraps and garden garbage
- Polyethylene
- Linen and carton
- Textiles
- Non-flammable minerals and glassware

Each of these categories has a different carbon emission factor, especially under different treatment processes (e.g., landfilling vs. incineration). The fraction of degradable organic carbon (DOC) is particularly important for landfill emission calculations. The emissions are typically estimated:

- Annually (e.g., tonnes CO_2e/year)
- By location/region (municipality-wise or plant-wise)

This allows policymakers to benchmark and track emissions over time and evaluate mitigation strategies such as:

- Enhancing landfill gas capture
- Promoting composting and recycling
- Reducing waste at source

Carbon emissions vary significantly based on the method of disposal or treatment, such as landfilling (with or without gas capture), incineration (with or without energy recovery), composting or anaerobic digestion and recycling. Each treatment method has an associated emission factor per tonne of waste, typically expressed in kg CO_2-equivalent (CO_2e)/tonne. The intricacy of SW production patterns is frequently not adequately captured by traditional forecasting techniques. Algorithmic methods for have become increasingly effective in forecasting garbage generation in recent years[24]. Because of its adaptability and resilience while working with complicated datasets, Random Forest has become well-known among these algorithms. In SW forecasting studies, Table 5 examines the controls for tuning and particular values for different machine learning algorithms, emphasising their advantages and their influence on environmentally friendly waste management techniques.

Machine Learning Algorithms	Tuning parameters	Experimented Values
Random Forest Algorithm	Hyperparameters undertaken to optimize model performance: **Trees Count** (n_estimators) **Highest Tree Depth** (max_depth) **Min_samples_split, or Minimum Samples Split** **Min_samples_leaf, or Minimum Samples Leaf** **Max_features, or Maximum Features** **State of Randomness** (random_state) **Weight of Class** (class_weight)	**(n_estimators)**: 4 **(max_depth)**: 8 **(min_samples_split)**: 10:100 **(min_samples_leaf)**: 2 **(max_features)**: 100 **(random_state)**: 2 **(class_weight)** :85
Navie Bayes	**Alpha (alpha)**: Controls additive smoothing helps in handling features with zero frequencies **Binarize** (binarize) **Fit Prior** (fit_prior)	**Alpha (alpha)**:0.75 **Binarize (binarize)**: 2 levels **Fit Prior (fit_prior)**: 95%
Support Vector Machine	**Cache Size** (cache_size) Choosing the Right Kernel **Probability Estimates** (probability) **Maximum Iterations** (max_iter) **Tolerance** (tol) **Coef0** (coef0)	90% Radial Basis Function (RBF) Cross Validation type 10:90 0.5 Sigmoid

Table 5. Specific machine learning parameters.

Data pre-processing: Data preparation is crucial before using ML algorithms on the available datasets. This entails addressing missing numbers, encoding category variables, and cleaning and processing raw SW data. The algorithm's ability to effectively extract important trends from the data is ensured by appropriate pre-processing.

Model training: Using historical SW generation data, the ML algorithm is trained to understand the connections between different factors and waste output. In order to ensure that the model can generalise to new data, cross-validation approaches aid in optimising hyperparameters[25].

Data quality: The quality of the input data has a significant impact on the precision and dependability of waste forecasting models. The ML model is expected to function successfully if the training data is typical of the real SW generation trends in India.

Magnitude and Local Alternatives: Waste generation patterns vary significantly by area in India, a large and diverse nation. It's critical to scale the model appropriately and take these regional variations into account. A model may not generalise effectively to other regions if it performs well in one.

Random forest (RF) algorithm in SW Estimating:Based on input factors, the Random Forest framework can be trained to predict future SW generation (Zhou et al. 2015). In order to evaluate the clustering standards in the context of estimating emissions along with improved optimization, the significant number of clusters are chosen. In current scenario, for unsupervised learning strategies, assessment is made using the identification of silhouette width[27]. Based on various locations, for each location p_n, the silhouette width is calculated using the following equation (6.1).

$$Q(p_n) = r(p_n) - s(p_n)/max(s(p_n), r(p_n)) \tag{3}$$

where,

- $s(p_n)$ is the average distance between the independent variables placed at p_n and all other samples.
- $r(p_n)$ is the minimum of the average distances between the pattern at location**Methods for SW Forecasting Using Naive Bayes (NB) and Support Vector Machines (SVM):** Based on Bayes' theorem, the approach known as Naive Bayes presumes that predictor variables are independent of one another. It is essential to select the right characteristics for SW forecasting. Density of population, financial indicators, rates of urbanisation, and other variables that affect garbage generation may be pertinent characteristics[26]. For the NB model to succeed, appropriate feature construction is crucial. The algorithm is the population density, indicates urbanization rates with rate of data duplication.

$$Q(x) = \theta_0 + \sum_{\mu=1}^{\infty}\left(a\rho_{r+1}cos\frac{v\Delta\theta}{\mu} + b_r sin\frac{v\Delta\theta}{\mu}\right) \tag{4}$$

For best results, SVM models' hyperparameters must be adjusted. The accuracy of the model can be affected by the regularisation parameter, kernel function, and other factors. It is crucial to adjust these parameters in accordance with the features of the waste data from India. SVM models must take temporal trends into consideration because waste creation patterns can change over time. The modelling approach should take into account time-dependent variables such as economic changes and seasonal variations. Careful evaluation of the validity of the data, picking features, emulate adjustments, seasonal issues, geographic variation, and contrasts with alternative approaches are all necessary for the efficacy of machine learning algorithms for SW foreseeing in the Indian setting (Zaeimi et al. 2021). Before using the model for real-world waste projection applications, extensive validation and testing are advised to evaluate its performance. It is clear that the range, mean, and average deviation for weighted components are calculated correctly; the datasets highest deviation was 3.058, therefore proximity ranking approaches must be used to lessen the datasets randomness.

	Proximity Matrix		
	Euclidean Distance		
	1: Paper and Cardboard Generated per zone	2: Zonal Plastic Waste Accumulated	3: Zonal Organic Waste Accumulated
1: Paper and Cardboard Generated per zone	.000	6097.000	3481.000
2: Zonal Plastic Waste Accumulated	6097.000	.000	2616.000
3: Zonal Organic Waste Accumulated	3481.000	2616.000	.000

Table 6. Proximity matrix of zonal waste datasets.

ML Models	Training MSE	Testing MSE
Number of Features =3		
SVM	6692	6856
RF	9027	10353
NB	5688	7665
Number of Features = 2 Excluding Category 2		
SVM	5431	4852
RF	6842	5461
NB	5237	5031

Table 7. Feature selection based on waste survey.

Proximity matrices are essential for understanding relationships between data points, facilitating tasks like clustering, anomaly detection, recommendation, and nearest-neighbour methods. Ranking is critical for tasks that require prioritization of results, such as search engines, recommender systems, and classification tasks with ordered outputs. Together, proximity matrices help define relationships, while ranking helps prioritize or order the results based on relevance or similarity in ML executions. The data mining is initiated with datasets holding paper and cardboard waste, zonal plastic waste and zonal organic waste, proximity ranking is executed to examine the ranking among these three influences on the waste predicting tuning model. Table 6 provides the proximity matrix generated for the given datasets of wastes undertaken at the zonal divisions.

This proximity matrix, which sheds a little understanding on the organisational framework and the distance attributes, is derived by taking into account the waste survey parameters and proximity to the collection system. The smallest distance in the matrix is between Plastic waste and Organic waste, indicating that their quantities in different zones are more similar to each other compared to their relationship with Paper and Cardboard waste. This could imply that zones accumulating more Plastic waste tend to also accumulate a comparable amount of Organic waste. Zones that generate a lot of Plastic waste tend to generate similar amounts of Organic waste, but these zones may produce significantly less Paper and Cardboard waste.

The important features were first chosen through trial and error. Then, using real-time data such as rubbish, silt quantity, and waste elements, polyester and food scraps were assigned weights for additional prediction, as indicated in Table 7. In the case of small datasets, it is important to depict the process of training and testing errors. Figure 10 displays the variation in the mean of error rates that suggests the model stability. The standard deviation for testing error does not exhibit an increasing trend, indicating that the model is subject to complex phenomena. This indicates that the model works well with the data.

The methodology described in this study shows that it is feasible to acquire, pre-process, integrate, and model readily accessible information from various sources in order to create tools for regional SW planning. To precisely forecast SW generation trends, three distinct machine learning techniques are used. To improve the models' accuracy, the training and testing stages are repeated.

Using WEKA software (version 3.8.5) statistics, the datasets were trained and tested using SVM, NB and RF algorithms. Figure 11 shows the accuracy of each model against the number of conducted test with variation tuning parameters. It is confirmed that the SVM exploits better accuracy in higher rate of 96% as compared with RF and NB algorithms.

Once the socio-economic parameters were selected for the model, the data partition ratio was adjusted to identify the optimal way to split the data. Table8 shows the variations in training and testing errors for different partition ratios, with the error decreasing as the size of the testing set increased, stabilizing at an 80:20 split. The model's complexity also influences its ability to learn underlying patterns, as determined by the machine learning algorithm. The computational time required by ML algorithms to process the wastes was investigated; all of the classifiers could operate on an Intel i5 processor with 8 GB of RAM. SVM was discovered to have the best accuracy in forecasting and the shortest computation time (0.67), which qualifies it for real-time applications. Combining statistical analysis and domain-dependent environmental implications inform the selection of the most influential waste kinds, such as polyethylene and vegetable (food/kitchen) waste, to fine-tune the learning machine (ML) model. According to the study's Tables 4 and 6 closeness matrix computation and geographical waste breakdown investigations verified that plastic and organic garbage were the main contributors in each of Erode City's zones. These two waste classes are more indicative of overall waste behaviour since they have the

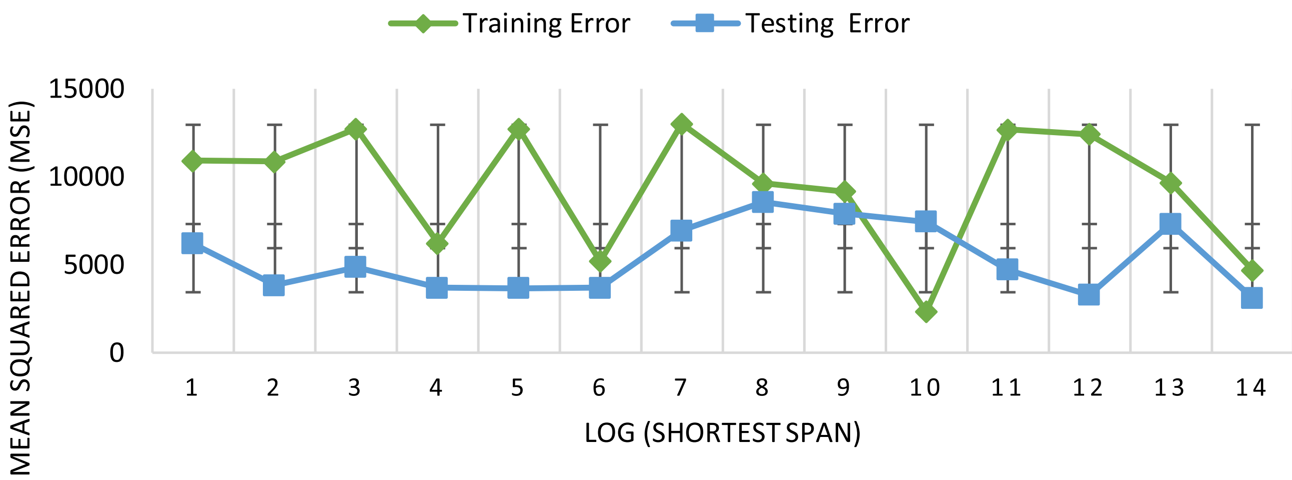

Fig. 10. Model average training and testing error.

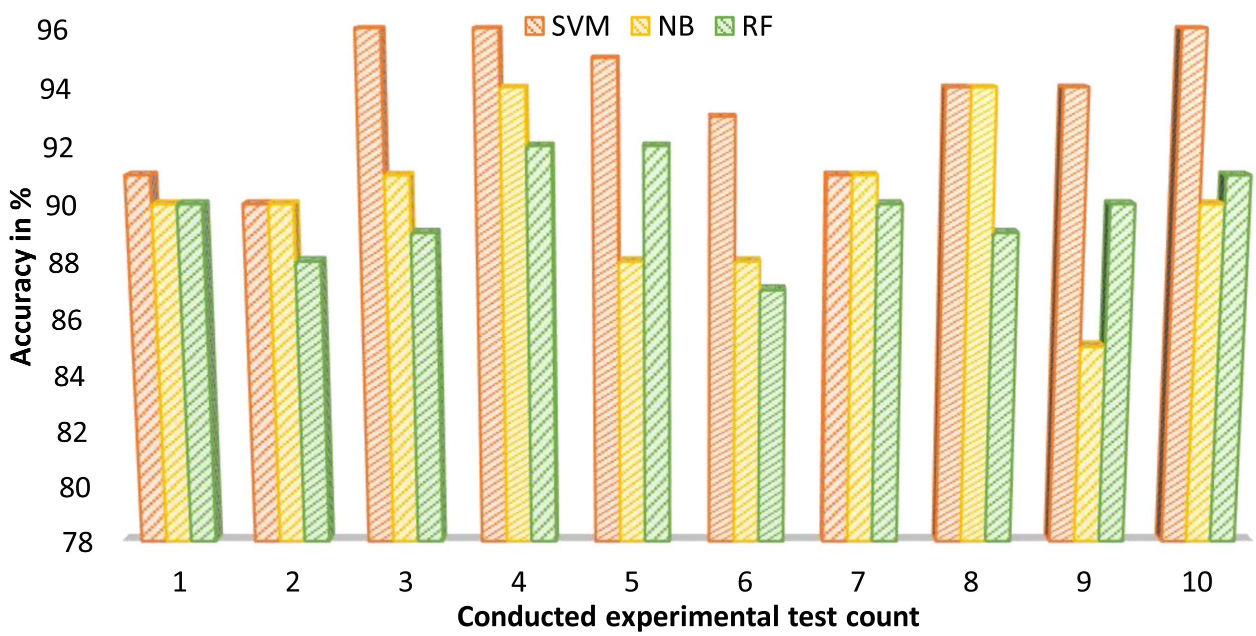

Fig.11. Model accuracy rates comparison analysis on waste forecasting.

Model	Accuracy	Model Computation Time	Error Rate
SVM	96%	0.67 Sec	Low
RF	91.54%	0.79Sec	Moderate
NB	93.21%	0.93Sec	High

Table 8. Efficiency comparison among Machine Learning algorithms.

highest correlation and the smallest Euclidean distance, suggesting that their quantities change similarly across regions. Enhancing the performance of the model, clarity, and pollution predictability are the main goals of using these contributing waste kinds in model tuning. In landfills, organic waste breaks down anaerobically, producing the powerful greenhouse gas methane (CH_4). When plastic waste is burned or improperly disposed of, it greatly increases CO2 emissions and long-term pollution. By concentrating on such SW, the model better fits the calculation of carbon footprints, which is the main objective of the suggested research project.

Recommendation:

- Implement Smart Bin Systems: Install IoT-enabled smart bins with sensors to monitor fill levels, emissions, and temperature, allowing dynamic routing and real-time waste management responses.
- Mandate Source-Level Waste Segregation: Promote household and institutional segregation of waste into biodegradable, recyclable, and non-recyclable categories through regulations and reward-based programs.
- Strengthen Municipal Data Infrastructure: Create a centralized and regularly updated database capturing socio-economic, demographic, and waste type information to support accurate machine learning predictions.
- Include Carbon Emission Metrics in Waste Planning: Assess greenhouse gas emissions from landfills and waste treatment facilities to inform decisions on sustainable disposal methods and climate compliance.
- Use Zonal Waste Profiles for Infrastructure Planning: Apply proximity ranking and feature selection to identify zones generating high plastic or organic waste, and allocate treatment units accordingly.
- Encourage Community Engagement via Digital Platforms: Develop mobile apps and web tools to involve the public in reporting waste-related issues, accessing collection schedules, and receiving segregation tips.
- Conduct Regular Field Surveys and Impact Studies: Undertake periodic surveys to assess the effectiveness of collection, segregation, and public awareness, and to recalibrate machine learning models accordingly.

Conclusions and future scope

The experimental analysis revealed a predictive framework for municipal solid waste (MSW) generation and emission analysis in Erode City using machine learning algorithms Support Vector Machine (SVM), Random Forest (RF), and Naive Bayes (NB). Among these, the SVM model demonstrated Superior performance with a prediction accuracy of 96% and the lowest testing Mean Squared Error (MSE) of 4860, indicating strong reliability and reduced error frequency. The RF and NB models achieved 91.5% and 93.2% accuracy respectively but showed higher error margins. Proximity ranking and feature selection methods identified plastic and organic wastes as the most influential contributors, with a Euclidean distance of 2616, Suggesting similar distribution patterns across city zones. Waste volume trends between 2019 and 2024 revealed a rise in garbage from 2566.65 MT/day to 3724.18 MT/day, stressing the growing demand for systematic interventions. The integration of ML with socio-economic and spatial indicators allows for efficient zonal planning, emission forecasting, and targeted infrastructure deployment. With a processing latency of just 0.67 seconds, the SVM model proves ideal for real-time applications. This research supports data-driven waste management strategies and encourages AI adoption for achieving long-term environmental sustainability.

Future scope:

- Develop superior waste generation models by refining data gathering and interpretation methods in order to apply improved solid waste management procedures.
- There is a great chance of achieving the zero waste phenomena on a national and international scale by integrating advanced technologies like deep machine learning, intelligent smart bins, and networks into the solid waste management process.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Additional information
Correspondence and requests for materials should be addressed to R.A. or B.K.